



United International University

Department of Computer Science and Engineering

CSE 3313: Computer Architecture

Midterm Examination: Summer 2025

Total Marks: 30 Time: 1 hour and 30 minutes

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. Consider computer A with a CPU speed of 2 GHz. The program P which has number of instruction and CPI as shown in Table 1.

Instruction	Count	CPI
addi	4	2
add	4	1
beq	2	4
bne	1	4
slt	2	3
sll	2	4
and	3	2
j	2	8

Table 1: Program P's number of instructions and CPI in Computer A architecture

Consider that Program P requires 50 ns to be completed Computer A.

- (i) Find the execution time of program P in Computer A. [2] **Solution 1 (i)**
Total Clock Pulse = $(4 \times 2) + (4 \times 1) + (2 \times 4) + (1 \times 4) + (2 \times 3) + (2 \times 4) + (3 \times 2) + (2 \times 8) = 60$ cycles
Clock Rate = 2×10^9 Hz
CPU Time = $60 / 2 \times 10^9 = 30 \times 10^{-9} \text{s} = 30 \text{ ns}$
- (ii) Computer B with a CPU speed of 10 GHz has the execution time of the program P which is now 2 times faster than computer A and same instruction set architecture (ISA), calculate the average CPI of computer B. [2] **Solution 1 (ii)**
Execution time on B = $30 \text{ ns} / 2 = 15 \text{ ns}$
Clock Rate = 10×10^9 Hz
Total Instructions = 20
Total Clock Pulse = Execution time $\times$ Clock Rate = $15 \text{ ns} \times 10 \times 10^9 \text{ Hz} = 150$
Average CPI = Total Clock Pulse / Total Instruction = $150 / 20 = 7.5$
- (iii) As we get the total execution time of the given program for Computer A from question (i), assume that the Arithmetic unit takes 65%, the logical unit takes 25%, and the branch operation takes 10% time of total execution time. Find the improvement factor of the given program if we replace the arithmetic unit with a better Arithmetic unit, which improves total completion time by 2x. [6] **Solution for 1 (iii)**
Given total completion time for program P on A is 50 ns and execution time is 30 ns from question (i)
Arithmetic: 65% of 30 ns = 19.5 ns
Logical: 25% of 30 ns = 7.5 ns
Branch: 10% of 30 ns = 3 ns
After improving total completion time by 2x:
Improved time = $50 / 2 = 25 \text{ ns}$
Affected time = 19.5 ns
Unaffected time = $7.5 + 3 + 20(\text{ system}) = 30.5$
so, $25 = 30.5 + 19.5/n$
2x improvement not possible, even if we replace the arithmetic unit with a better one
2. (a) Consider the following C function. The arguments will be found at \$a0-\$a2. The compiler will assign a at \$s0. Convert the code to the corresponding MIPS assembly instructions. [8]

```

1  int my_function(int i, int n, int arr[])
2  {
3      int a = 10;
4      for(i = 0; i <= n; i++)
5      {
6          arr[ i * 2 ] = ~ (a & n);
7      }
8      return a;
9  }

```

Code for 2 (a)

Solution for 2 (a) :

```

1  my_function:
2  addi $sp, $sp, -4
3  sw $s0, 0($sp)
4  addi $s0, $zero, 10
5  LOOP:
6      slt $t0, $a1, $a0
7      beq $t0, $zero, REST
8      sll $t1, $s0, 3
9      addi $t2, $t1, $a2
10     and $t3, $s0, $a1
11     nor $t3, $zero, $t3
12     sw $t3, 0 ( $t2)
13     addi $a0, $a0, 1
14     j LOOP
15     REST:
16     addi $v0, $zero, $s0
17     lw $s0, 0($sp)
18     addi $sp, $sp, 4
19     jr $ra

```

- (b) Observe the following MIPS code and convert it into machine code. Conversion of the machine code to binary is not required. [3]

```

1  100: LOOP
2  104: slt $t3, $t5, $s1
3  108: bne $t3, $zero, REST
4  112: sw $t5, 20 ( $s4)
5  116: j LOOP
6  120: andi $t2, $s3, 30
7  124: REST
8  128: sub $t2, $a0, $s3

```

Code for 2 (b)

Solution for 2 (b)

i. slt \$t3, \$t5, \$s1

Opcode	Rs	Rt	Rd	Shamt	Function Code
0	13	17	11	0	42

ii. bne \$t3, \$zero, REST

Opcode	Rs	Rt	Constant or Address
5	11	0	31

iii. sw \$t5, 20 (\$s4)

iv. j LOOP

v. andi \$t2, \$s3, 30

Opcode	Rs	Rt	Constant or Address
43	20	13	20

Opcode	Address
2	25

Opcode	Rs	Rt	Constant or Address
12	19	10	30

Opcode	Rs	Rt	Rd	Shamt	Function Code
0	4	19	10	0	34

vi. sub \$t2, \$a0, \$s3

- (c) The students of UIU have designed a modified instruction set architecture called 'MIPS_3313' that follows the principles of the MIPS architecture but only with a few changes. The processor of MIPS_3313 has **16 registers**. The architecture is **byte-addressable** and the size of the instruction and register is **16 bits**. **3** bits are reserved for opcode. The structure for lw instruction in MIPS_3313 is given in the table-2. **Calculate** the maximum size of the array that the MIPS_3313 instruction can operate with.. [4]

Solution for 2 (c)

Processor has 16 registers $\rightarrow$ each register field requires 4 bits (since $2^4 = 16$)

Instruction size: 16 bits

Opcode bits = 3

Bits for rs = 4

Bits for rt = 4

So, bits for offset = $16 - (3 + 4 + 4) = 5$ bits

Maximum Size for constant field = $2^5 - 1 = 31$

Maximum Index = $31/2 = 15$

Maximum Size = 16

3. (a) Multiply two numbers X by Y using the optimized multiplication algorithm, where X = 5 and Y = 6. [4]

Solution for 3 (a)

5 = 0101 $\rightarrow$ multiplicand (mN)

6 = 0110 $\rightarrow$ multiplier (mR)

30 = 11110 $\rightarrow$ product (P)

P: [0000 0110]

Step 1:

P0 = 0

P: [0000 0011]

Step 2:

P0 = 1

MN: 0101 + PLH (0000) = 0101

P: [0101 0011]

Right Shift

P: [0010 1001]

Step 3:

P0 = 1

MN: 0101 + PLH (0010) = 0111

P: [0111 1001]

Right Shift

P: [0011 1100]

opcode	rs	rt	constant
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Table 2: Structure of lw instruction

Step 4:

P0 = 0

P: [0001 1110]

Final Answer: 11110

- (b) You need to compute $P=A \times B \times C$ using unsigned binary integers. Let A be 9 bits wide, B be 14 bits wide, and C is defined as $C=A \times B$. What is the minimum width (in bits) required for the product register to hold P at any time? [1]

Solution for 3 (b)

$C= A \times B = (9 + 14)$ bits

C requires 23 bits

$P = A \times B \times C = C \times C (23 + 23) = 46$ bits

P Total Requires 64 bits (power of 2)

Instruction	Opcode	Function Code
add	0	32
sub	0	34
lw	35	-
sw	43	-
and	0	36
or	0	37
nor	0	39
andi	12	-
ori	13	-
sll	0	0
srl	0	2
beq	4	-
bne	5	-
slt	0	42
j	2	-
jr	0	8
jal	3	-
addi	8	-

(a) MIPS Machine Codes

Name	Register Number
\$zero	0
\$at	1
\$v0-\$v1	2-3
\$a0-\$a3	4-7
\$t0-\$t7	8-15
\$s0-\$s7	16-23
\$t8-\$t9	24-25
\$k0-\$k1	26-27
\$gp	28
\$sp	29
\$fp	30
\$ra	31

(b) MIPS Registers